behavior to make the effort more manageable [17, 73]. Rule-based approaches, such as finite-state machines [91, 97] and behavior trees [41, 54, 82] account for the brute force approach of human-authoring the agent's behavior [71]. They provide a straightforward way of creating simple agents that is still the most dominant approach today [69, 74, 108], and can even handle rudimentary social interactions, as shown in games such as Mass Effect [13] and The Sims [7] series. Nonetheless, manually crafting behavior that can comprehensively address the breadth of possible interactions in an open world is untenable. This means that the resulting agent behaviors may not fully represent the consequences of their interactions [70–72], and cannot perform new procedures that were not hard-coded in their script [91, 97]. On the other hand, prevalent learning-based approaches for creating believable agents, such as reinforcement learning, have overcome the challenge of manual authoring by letting the agents learn their behavior, and have achieved superhuman performance in recent years in games such as AlphaStar for Starcraft [99] and OpenAI Five for Dota 2 [11]. However, their success has largely taken place in adversarial games with readily definable rewards that a learning algorithm can optimize for. They have not yet addressed the challenge of creating believable agents in an open world [40, 74, 91].

Cognitive architectures in computation, pioneered by Newell, aimed to build the infrastructure for supporting a comprehensive set of cognitive functions [76] that suited the all-encompassing nature of believable agents held in its original vision. They fueled some of the earliest examples of believable agents. For instance, Quakebot-SOAR [60] and ICARUS [25, 64] generated NPCs in first-person shooter games, while TacAir-SOAR [81] generated pilots in aerial combat training simulations. The architectures used by these agents differed (Quakebot- and TacAir-SOAR relied on SOAR [61], while ICARUS relied on its own variation that was inspired by SOAR and ACT-R [6]), but they shared the same underlying principle [62]. They maintained short-term and long-term memories, filled these memories with symbolic structures, and operated in perceive-plan-act cycles, dynamically perceiving the environment and matching it with one of the manually crafted action procedures [58, 97]. Agents created using cognitive architectures aimed to be generalizable to most, if not all, open world contexts and exhibited robust behavior for their time. However, their space of action was limited to manually crafted procedural knowledge, and they did not offer a mechanism through which the agents could be inspired to seek new behavior. As such, these agents were deployed mostly in non-open world contexts such as first-person shooter games [25, 60] or blocks worlds [64].

Today, creating believable agents as described in its original definition remains an open problem [85, 108]. Many have moved on, arguing that although current approaches for creating believable agents might be cumbersome and limited, they are good enough to support existing gameplay and interactions [24, 75, 108]. Our argument is that large language models offer an opportunity to re-examine these questions, provided that we can craft an effective architecture to synthesize memories into believable behavior. We offer a step toward such an architecture in this paper.

2.3 Large Language Models and Human Behavior

Generative agents leverage a large language model to power their behavior. The key observation is that large language models encode a wide range of human behavior from their training data [15, 18]. If prompted with a narrowly defined context, the models can be used to generate believable behavior. Recent work has demonstrated the efficacy of this approach. For instance, social simulacra used a large language model to generate users that would populate new social computing systems to prototype their emergent social dynamics [80]. This approach used a prompt chain [105, 106] to generate short natural language descriptions of personas and their behaviors as they appear in the system being prototyped. Other empirical studies have replicated existing social science studies [46], political surveys [92], and generated synthetic data [39]. Large language models have also been used to generate interactive human behavior for users to engage with. In gaming, for instance, these models have been employed to create interactive fiction [37] and text adventure games [21]. With their ability to generate and decompose action sequences, large language models have also been used in planning robotics tasks [48]. For example, when presented with a task, such as picking up a bottle, the model is prompted to break down the task into smaller action sequences, such as heading to the table where the bottle is located and picking it up.

We posit that, based on the work summarized above, large language models can become a key ingredient for creating believable agents. The existing literature largely relies on what could be considered first-order templates that employ few-shot prompts [38, 66] or chain-of-thought prompts [100]. These templates are effective in generating behavior that is conditioned solely on the agent's current environment (e.g., how would a troll respond to a given post, what actions would a robot need to take to enter a room given that there is a door). However, believable agents require conditioning not only on their current environment but also on a vast amount of past experience, which is a poor fit (and as of today, impossible due to the underlying models' limited context window) using first-order prompting. Recent studies have attempted to go beyond first-order prompting by augmenting language models with a static knowledge base and an information retrieval scheme [53] or with a simple summarization scheme [104]. This paper extends these ideas to craft an agent architecture that handles retrieval where past experience is dynamically updated at each time step and mixed with agents' current context and plans, which may either reinforce or contradict each other.

3 GENERATIVE AGENT BEHAVIOR AND INTERACTION

To illustrate the affordances of generative agents, we instantiate them as characters in a simple sandbox world reminiscent of The Sims [7]. This sprite-based sandbox game world, Smallville, evokes a small town environment. In this section, we will walk through the affordances and interactions with generative agents in Smallville and describe how the agents behave within it. Then, in Section 4, we will introduce our generative agent architecture that powers these affordances and interactions. In Section 5, we will describe the

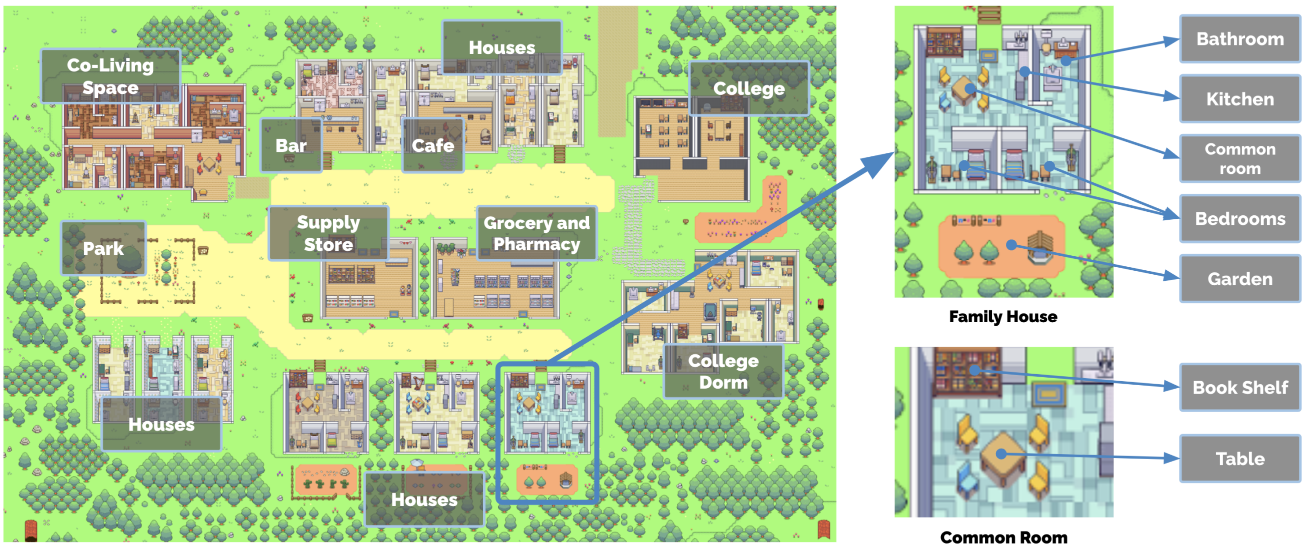

Figure 2: The Smallville sandbox world, with areas labeled. The root node describes the entire world, children describe areas (e.g., houses, cafe, stores), and leaf nodes describe objects (e.g., table, bookshelf). Agents remember a subgraph that reflects the parts of the world they have seen, maintaining the state of those parts as they observed them.

implementation of the sandbox environment and how the agents interact with the underlying engine of the sandbox world.

3.1 Agent Avatar and Communication

A community of 25 unique agents inhabits Smallville. Each agent is represented by a simple sprite avatar. We authored one paragraph of natural language description to depict each agent's identity, including their occupation and relationship with other agents, as seed memories. For example, John Lin has the following description:

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John Lin is a pharmacy shopkeeper at the Willow Market and Pharmacy who loves to help people. He is always looking for ways to make the process of getting medication easier for his customers; John Lin is living with his wife, Mei Lin, who is a college professor, and son, Eddy Lin, who is a student studying music theory; John Lin loves his family very much; John Lin has known the old couple next-door, Sam Moore and Jennifer Moore, for a few years; John Lin thinks Sam Moore is a kind and nice man; John Lin knows his neighbor, Yuriko Yamamoto, well; John Lin knows of his neighbors, Tamara Taylor and Carmen Ortiz, but has not met them before; John Lin and Tom Moreno are colleagues at The Willows Market and Pharmacy; John Lin and Tom Moreno are friends and like to discuss local politics together; John Lin knows the Moreno family somewhat well — the husband Tom Moreno and the wife Jane Moreno.
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Each semicolon-delimited phrase is entered into the agent's initial memory as memories at the start of the simulation.

3.1.1 Inter-Agent Communication.

The agents interact with the world by their actions, and with each other through natural language. At each time step of the sandbox engine, the agents output a natural language statement describing their current action, such as "Isabella Rodriguez is writing in her journal", "Isabella Rodriguez is checking her emails", "Isabella Rodriguez is talking with her family on the phone", or "Isabella Rodriguez is getting ready for bed." This statement is then translated into concrete movements that affect the sandbox world. The action is displayed on the sandbox interface as a set of emojis, providing an abstract representation of the action from an overhead view. To achieve this, the system utilizes a language model to translate the action into a set of emojis, which appear above each avatar's head in a speech bubble. For example, "Isabella Rodriguez is writing in her journal" is displayed as 📖✏️, while "Isabella Rodriguez is checking her emails" appears as 💻✉️. The complete natural language description of the action can be accessed by clicking on the agent's avatar.

Agents communicate with each other in full natural language. They are aware of other agents in their local area, and the generative agent architecture determines whether they walk by or engage in conversation. Here, a sample in the middle of a conversation between the agents Isabella Rodriguez and Tom Moreno about the upcoming election:[3]

> Isabella: I'm still weighing my options, but I've been discussing the election with Sam Moore. What are your thoughts on him?
> Tom: To be honest, I don't like Sam Moore. I think he's out of touch with the community and doesn't have our best interests at heart.

[3]We note that the conversational style of these agents can feel overly formal, likely a result of instruction tuning in the underlying models. We expect that the writing style will be better controllable in future language models.

3.1.2 User Controls. The user communicates with the agent through natural language by specifying a persona that the agent should perceive them as. For example, if the user specifies that they are a news "reporter" and asks about the upcoming election by saying, "Who is running for office?", the John agent replies:

> John: My friends Yuriko, Tom and I have been talking about the upcoming election and discussing the candidate Sam Moore. We have all agreed to vote for him because we like his platform.

To directly command one of the agents, the user takes on the persona of the agent's "inner voice"—this makes the agent more likely to treat the statement as a directive. For instance, when told "You are going to run against Sam in the upcoming election" by a user as John's inner voice, John decides to run in the election and shares his candidacy with his wife and son.

3.2 Environmental Interaction

Smallville features the common affordances of a small village, including a cafe, bar, park, school, dorm, houses, and stores. It also defines subareas and objects that make those spaces functional, such as a kitchen in a house and a stove in the kitchen (Figure 2). All spaces serving as agents' primary living quarters feature a bed, desk, closet, shelf, as well as a bathroom and a kitchen.[4]

Agents move around Smallville as one would in a simple video game, entering and leaving buildings, navigating its map, and approaching other agents. Agent movements are directed by the generative agent architecture and the sandbox game engine: when the model dictates that the agent will move to a location, we calculate a walking path to the destination in the Smallville environment, and the agent begins moving. In addition, users can also enter the sandbox world of Smallville as an agent operating within it. The agent that the user embodies can be an agent already present in the world, such as Isabella and John, or it can be an outside visitor with no prior history in Smallville. The inhabitants of Smallville will treat the user-controlled agent no differently than they treat each other. They recognize its presence, initiate interactions, and remember its behavior before forming opinions about it.

Users and agents can influence the state of the objects in this world, much like in sandbox games such as The Sims. For example, a bed can be occupied when an agent is sleeping, and a refrigerator can be empty when an agent uses up the ingredients to make breakfast. End users can also reshape an agent's environment in Smallville by rewriting the status of objects surrounding the agent in natural language. For instance, when Isabella is making breakfast in the morning, the user can change the status of the kitchen stove from "turned on" to "burning" by inputting a command to the system that chooses the object and illustrates its new status, like this: "<Isabella's apartment: kitchen: stove> is burning." Isabella will notice this in the next moment and go to turn off the stove and remake her breakfast. Likewise, if the user sets the status of Isabella's shower to "leaking water" when she enters the bathroom, she will gather tools from her living room and try to fix the leak.

[4]This environment design is not the focus of our work, so we generated this environment manually, not automatically. Future work can continue to expand the richness of the agents' environments.

3.3 Example "Day in the Life"

Starting from the single-paragraph description, generative agents begin planning their days. As time passes in the sandbox world, their behaviors evolve as these agents interact with each other and the world, building memories and relationships, and coordinating joint activities.

We demonstrate the behavior of generative agents by tracing the output of our system over the course of one day for the agent John Lin (Figure 3). In the Lin family, John is the first to wake up at 7 a.m. He brushes his teeth, takes a shower, gets dressed, eats breakfast, and checks the news at the dining table in the living room. At 8 a.m., Eddy follows suit, rushing out of bed to prepare for his classes. He catches up with John just as he is about to head out:

> John: Good morning Eddy. Did you sleep well?
> Eddy: Good morning dad. Yeah, I slept great.
> John: That's good. What are you working on today?
> Eddy: I'm working on a new music composition for my class. It's due this week, so I'm trying to get it finished. But I'm having so much fun with it!
> John: That sounds great!

Soon after Eddy heads out, Mei wakes up and joins John. Mei asks about her son, and John recalls the conversation they just had:

> Mei: Did Eddy already leave for school?
> John: Yes, he just left. He's working on a music composition for his class.
> Mei: Oh, that's wonderful! Tell me about it.
> John: I think he's really enjoying it! He said he's having a lot of fun with it.
> Mei: That's great! I'm so proud of him.

As they finish their conversation, Mei and John pack up. Mei goes on to teach and work on her research paper, while John opens his pharmacy counter at the Willow Market and Pharmacy by 9 a.m.

3.4 Emergent Social Behaviors

By interacting with each other, generative agents in Smallville exchange information, form new relationships, and coordinate joint activities. Extending prior work [80], these social behaviors are emergent rather than pre-programmed.

3.4.1 Information Diffusion. As agents notice each other, they may engage in dialogue—as they do so, information can spread from agent to agent. For instance, in a conversation between Sam and Tom at the grocery store, Sam tells Tom about his candidacy in the local election:

> Sam: Hey Tom, how's it going?
> Tom: Good, thanks. What's up?
> Sam: Well, I wanted to talk to you about something. I'm actually running for mayor in the upcoming local election.
> Tom: Really? That's great news! Why are you running?
> Sam: I've been involved in local politics for years now, and I really feel like I can make a difference in our community. Plus, I want to bring new ideas and perspectives to the table.